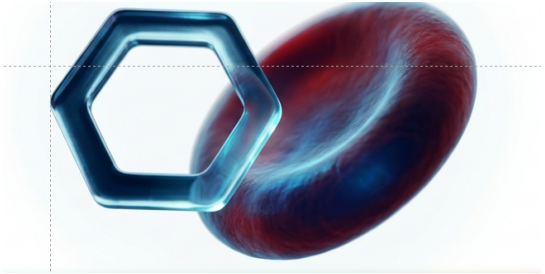
Hawler Medical University
College of Medicine

Determination of Serum Glucose

Biochemistry Lab 2
2026-2027
For 2nd-Stage Medical Student

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Learning Outcomes

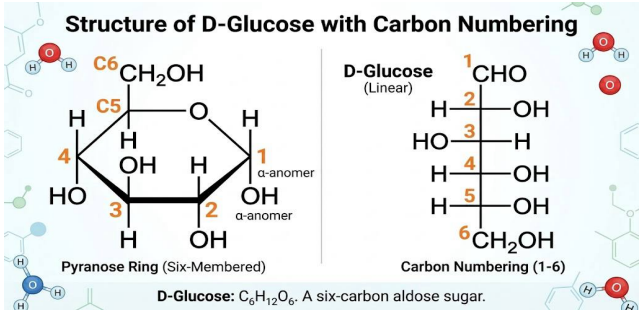
By the end of this session, students should be able to:

- Connect serum glucose measurement with glucose homeostasis and disease recognition.
- Compare the clinical patterns and causes of hyperglycemia and hypoglycemia.
- Summarize the GOD–POD colorimetric reaction used for glucose estimation.
- Explain why HbA1c represents longer-term, approximately three-month, glycemic exposure.
- Recognize the broad reference categories used for serum glucose and HbA1c interpretation.

Overview Glucose

Definition & Importance

- **Glucose** is a major circulatory carbohydrate and the **primary energy source** for the human body.
- As one of the three major classes of biological molecules, carbohydrates (CHO) are essential components of all living organisms.



Primary Sources of Blood Glucose

1. Dietary Intake

Carbohydrate digestion and absorption.

2. Body Stores

Glycogen breakdown in the liver and muscles.

3. Endogenous Synthesis

Gluconeogenesis from protein and glycerol.

Master Diabetes Diagnostic Standards

Diagnostic test	Normal	Prediabetes	Diabetes Mellitus
HbA1c (Glycated Hb)	<5.7	5.7%-6.4%	≥ 6.5
Fasting Glucose (FPG)	<100mg/dL (70-99 mg/dL)	100-125 mg/dL (IFG)	≥126mg/dL (confirmed)
2-Hour OGTT(75mg)	<140 mg/dL	140-199 mg/dL (IGT)	≥200 mg/dL (confirmed)
Random Glucose	70-140 mg/dL	Not Applicable N/A	≥ 200mg/dL (+Symptoms)

Clinical Significance: Hyperglycemia

Definition

Hyperglycemia refers to an abnormally **increased** concentration of glucose in the blood.

Diagnostic Value

It is a primary indicator for the diagnosis of **Diabetes Mellitus**.

Common Causes

- **Diabetes Mellitus:** Insulin resistance or absolute/ relative lack of insulin.
- **Pancreatic Disorders:** Acute or chronic pancreatitis.
- **Endocrine Disorders:** Cushing's syndrome and Hyperthyroidism.
- **Physiological Stress:** Burns, surgery, and trauma (increased cortisol/adrenaline).
- **Hormone-Secreting Tumors:** Glucagonoma or Pheochromocytoma.
- **Medications:** Estrogens, beta-blockers, and glucagon.

Clinical Significance: Hypoglycemia

Common Causes

- Excess insulin or glucose-lowering therapy
- Fasting, starvation, or malabsorption
- **Insulinoma** (pancreatic islet cell tumor)
- **Liver damage** (impaired glycogen storage)
- **Endocrine disorders** (Addison's, Hypopituitarism)

Insulin Mechanisms

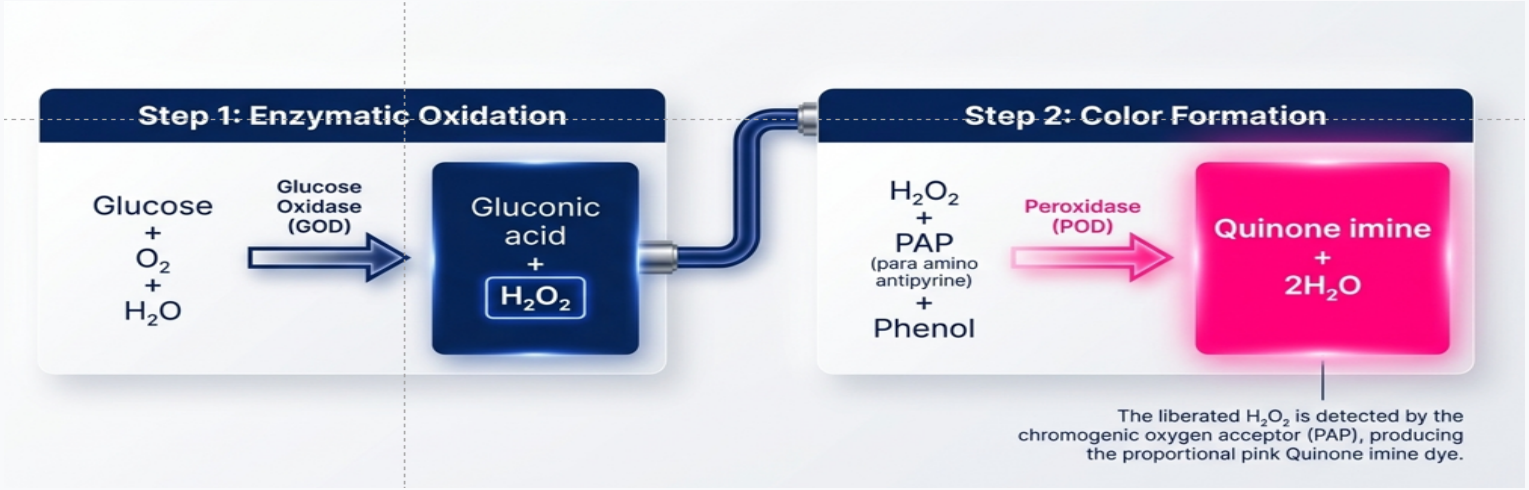
Promotes: Cellular glucose uptake, Glycolysis, and Glycogenesis.

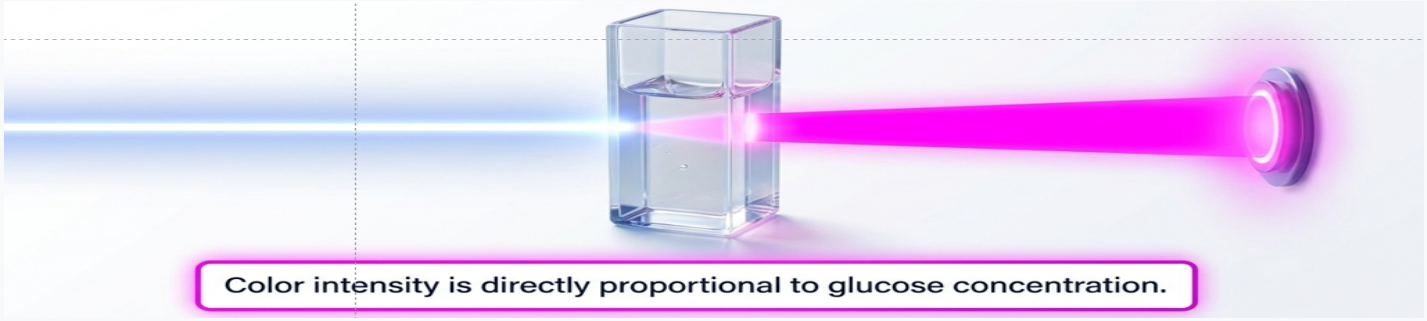
Inhibits: Glycogenolysis, Gluconeogenesis, and Lipolysis.

Result: *Rapid reduction in circulating blood glucose levels.*

Laboratory Principle

Glucose Oxidase–Peroxidase (GOD-POD) Method





Glycated Hemoglobin (HbA1c)

What is HbA1c?

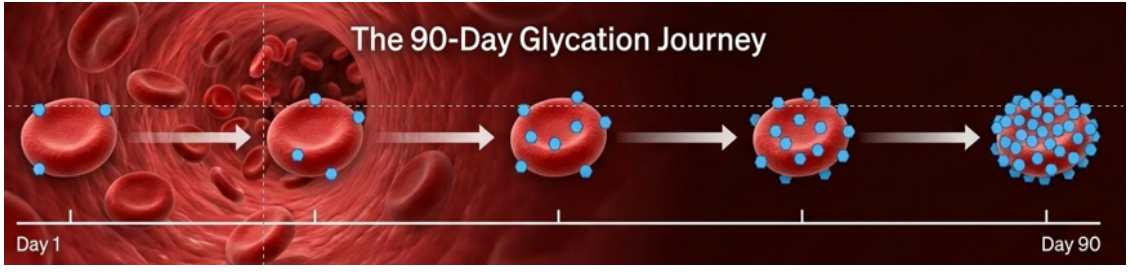
- HbA1c is a form of hemoglobin that measures the amount of blood glucose attached to hemoglobin over time.

The Three-Month Average

- Red blood cells typically live for about three months. This lifespan is the primary reason HbA1c provides a reliable long-term average.

"

- Stable Clinical Metric: Unlike daily glucose tests, HbA1c is not affected by recent food intake or short-term exercise.
  Diagnostic Window: Gives a comprehensive 2–3 month evaluation of overall blood glucose management.



HbA1c: Diagnostic Application

Reference Categories

Category	HbA1c Level
Normal	Less than 5.7%
Prediabetes	5.7% to 6.4%
Diabetes	6.5% or higher

Clinical Use

- **Screening:** Used to check for diabetes or prediabetes in adults.
- **Monitoring:** For individuals already diagnosed with diabetes, the test helps **monitor long-term** glucose control.

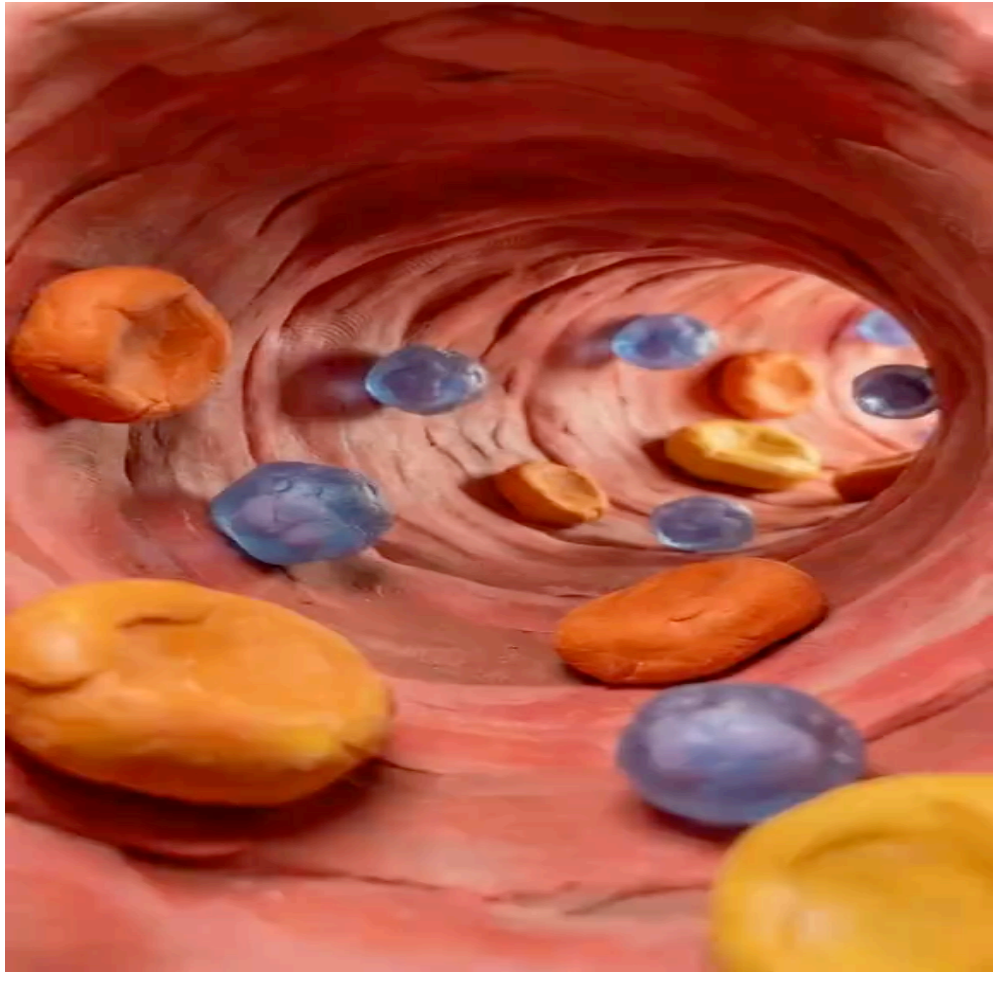
Note: High HbA1c levels indicate a higher risk of developing diabetes-related complications.

Insulin & Its Secretion

•**Insulin**: A hormone produced by the pancreas that **helps body cells absorb glucose** from the blood to use for **energy**.

• **Insulin Secretion:** The process where **beta** cells in the **pancreas** release insulin into the bloodstream, triggered primarily by a **rise in** blood glucose after eating.

🔑 How does insulin work in the body?



How Does a Glucometer Work?

1. Blood sample
A **small drop of blood** is placed **on a test strip**.

2. Enzyme reaction
Glucose reacts with an **enzyme on the strip**, such as **glucose oxidase or glucose dehydrogenase**.

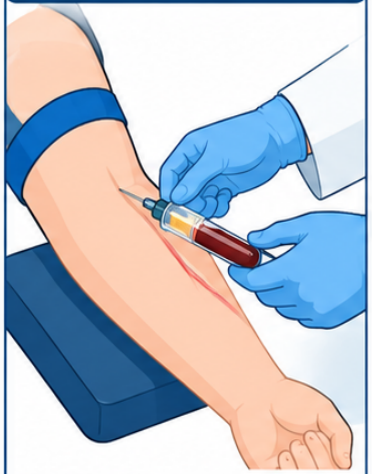
3. Electrical signal
The reaction produces an **electrical signal** that depends on the amount of glucose.

4. Glucose reading
The meter measures the signal, **calculates the glucose concentration**, and displays the result.



Serum Preparation from Whole Blood

1. Venous blood collection



Collect venous blood using a vacutainer.

2. Collect whole blood



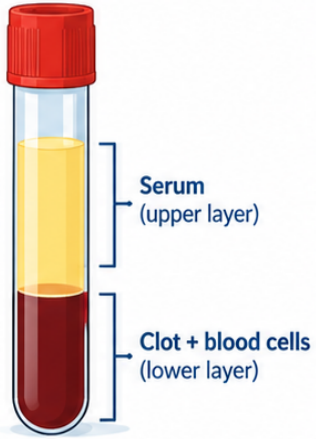
Whole blood collected in a plain (red-top) tube.

3. Centrifuge



Centrifuge the clotted blood (per your laboratory SOP).

4. Serum separation



After centrifugation, serum is separated from the clot.

5. Transfer serum



Carefully transfer the clear serum into a clean, labeled tube.

For serum: allow blood to clot before centrifugation.

Execution protocol and mathematical calculation for serum glucose.

1. Pipetting Scheme

Reagents	Blank	Standard	Sample
Glucose Standard (S)	-	10 µL	-
Sample (Serum/Plasma)	-	-	10 µL
Reagent (A)	1.0 mL	1.0 mL	1.0 mL

2. Test Tube Setup



3. Calculation Formula

$$\text{Glucose (mg/dL)} = \frac{A_{Sample}}{A_{Standard}} \times 100$$

A_{Sample} = Absorbance of Sample	$A_{Standard}$ = Absorbance of Standard
100 = Standard Conc. (mg/dL)	

- **Incubation:** Mix thoroughly & incubate for **5 min at 37°C** *(or 10 min at 16–25°C)*.
- **Absorbance:** Read absorbance (A) at **479 nm** against Blank within 2 hours

References

01 **TEXTBOOK**

Kennelly, P. J., Botham, K. M., McGuinness, O. P., Rodwell, V. W., & Weil, P. A. *Harper's Illustrated Biochemistry*, 32nd ed. McGraw Hill.

https://accessmedicine.mhmedical.com/book.aspx?bookid=3220

02 **CURRENT ARTICLE**

American Diabetes Association Professional Practice Committee. "2. Diagnosis and Classification of Diabetes: Standards of Care in Diabetes—2026." *Diabetes Care*. 2026;49(Supplement 1):S27–S49.

https://diabetesjournals.org/care/article/49/Supplement_1/S27/163926/2-Diagnosis-and-Classification-of-Diabetes

These sources support the biochemical background, diagnostic interpretation, and HbA1c discussion presented in this deck.

THANK YOU FOR YOUR TIME. MAY OUR SHARED KNOWLEDGE INSPIRE FUTURE INNOVATION

Thank You

Questions & Discussion

